Introduction to ML: Past and Present (2026)

ES 335 / ES 678 Machine Learning, Sem I 2026 — Prof. Udit Bhatia, Prof. Krishna Prasad Miyapuram Source: `Lectures/L01_student_ml_intro_past_present_2026.pdf` (26 slides) · Next: Mathematical Foundations

> **How to use this file.** Part A builds every bit of maths Lecture 1 uses, starting from Class 10 (algebra, coordinate geometry, quadratics, basic probability). Part B walks through all 26 slides and links each idea back to Part A. Part C is practice. Strong readers can start at Part B and jump back to Part A when a symbol is unfamiliar.

The most important mindset: **mathematical notation is compression.** Every symbol is a short name for an idea you could say in a sentence. Reading a formula means decompressing it back into that sentence.

Guiding question of the lecture: *When a model looks impressive, what would make the result untrustworthy?*

Step-up path

You need. Arithmetic and curiosity. Part A rebuilds everything else from the start.

1. Think of a model as a machine with knobs, and learning as turning them.
2. Measure wrongness with a loss, and average it to get the empirical risk $\hat{R}$.
3. Separate $\hat{R}$, which you can compute, from the true risk R, which you actually care about.
4. Use the four paradoxes as a checklist of how a good-looking number can mislead.

You can stop when you can say why a 99% accurate model might be worthless.

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Part A — Maths from the ground up

1. Sets · 2. Indices, hat/star · 3. Functions · 4. Summation · 5. Weighted averages · 6. Vectors · 7. Slopes & trends · 8. Probability · 9. Random variables & expectation · 10. Exponents, e, logs · 11. Loss building blocks · 12. min / argmin · 13. Confusion matrix · 14. Curve fitting & capacity · 15. Formula decoder

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Part A Maths From the Ground Up

1 Sets — the containers

Intuition. A set is a bag of distinct things. Order doesn't matter, duplicates don't count.

Notation.

Write	Say	Means
$A = \{1, 2, 3\}$	"A is the set 1, 2, 3"	listing members
$2 \in A$	"2 is in A"	membership
$5 \notin A$	"5 is not in A"	
$B \subseteq A$	"B is a subset of A"	everything in B is also in A
$\{x \in \mathbb{R} : x > 0\}$	"all real x such that x > 0"	*set-builder*: describe members by a rule
$\lvert A \rvert$	"size of A"	number of members (here 3)

The number sets (you met these in class 9–10):

$$\mathbb{N} = \{1, 2, 3, \dots\} \subseteq \mathbb{Z} = \{\dots, -1, 0, 1, \dots\} \subseteq \mathbb{Q}\ (\text{fractions}) \subseteq \mathbb{R}\ (\text{all points on the number line})$$

ML lives almost entirely in $\mathbb{R}$.

Pairs and the Cartesian product. An *ordered pair* (a, b) cares about order: $(2, 5) \neq (5, 2)$. The set of all pairs is

$$A \times B = \{(a, b) : a \in A,\ b \in B\}.$$

You already know one: the coordinate plane is $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$. Every point (x, y) is a pair of reals.

Generalise: $\mathbb{R}^3$ = all triples (3D space), and $\mathbb{R}^d$ = all lists of d real numbers. You can't *picture* $\mathbb{R}^{100}$, but you can *compute* in it exactly as in $\mathbb{R}^2$ — that's the trick ML uses constantly.

Why ML needs this. A training example is a pair (x, y): the input and its correct answer. A flood event described by (rainfall, area, slope) is $x \in \mathbb{R}^3$. A yes/no label is $y \in \{0, 1\}$. A whole dataset is a set of such

pairs.

Check: What is $\{0,1\} \times \{0,1\}$? → $\{(0,0),(0,1),(1,0),(1,1)\}$, four pairs.

2 Indices, subscripts, and the hat/star convention

Subscripts count. When you have many similar things, number them: $x_1, x_2, \ldots, x_n$. The letter i is the conventional "which one" counter, so x_i means "the i-th input" and (x_i, y_i) means "the i-th example".

Standard letters in ML (memorise these; they're used everywhere):

Letter	Meaning
n	number of examples (rows in the spreadsheet)
d	number of features (columns)
i	index over examples, $i = 1, \ldots, n$
j	index over features, $j = 1, \ldots, d$
x_{ij}	feature j of example i (one cell of the spreadsheet)
θ (theta)	the model's adjustable numbers (parameters)
w, b	weights and bias — the parameters of a linear model

Decorations carry meaning:

- **Hat** $\hat{y}$, $\hat{R}$, $\hat{f}$ → *estimated / computed from data*. "My guess."
- **Star** $f^\star$ → *the true, unknown thing*. "Nature's answer."
- **No decoration** → the general object or the true quantity.

So y is the real flood depth and $\hat{y}$ is what the model predicted. R is the true error and $\hat{R}$ is the error we measured on our data. **The whole of ML is the gap between hat and no-hat.**

Collections with indices:

$$\{(x_i, y_i)\}_{i=1}^{n} \quad = \quad \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$$

Read: "the pairs (x_i, y_i) for i running from 1 to n."

3 Functions — machines with knobs

Intuition. A function is a machine: put an input in, get **exactly one** output out. Same input → always same output.

Notation.

$$f : \mathcal{X} \to \mathcal{Y}$$

"f takes inputs from the set $\mathcal{X}$ (the *domain*) and gives outputs in the set $\mathcal{Y}$ (the *codomain*)." Then $f(x)$ is the output for input x.

Example	Domain → Codomain
$f(x) = x^2$	$\mathbb{R} \to \mathbb{R}$
"is this email spam?"	emails $\to \{0, 1\}$
"predict flood depth from 3 features"	$\mathbb{R}^3 \to \mathbb{R}$
loss $\ell(\hat{y}, y)$ — takes *two* numbers	$\mathbb{R} \times \mathbb{R} \to \mathbb{R}$

Functions with knobs (parameters). From class 10 you know $y = mx + c$. That's not *one* line; it's a *family* of lines. Each choice of (m, c) picks one. ML renames:

$$f_\theta(x) = wx + b, \qquad \theta = (w, b).$$

- x is the **input** — changes for every example.
- θ is the **knob setting** — fixed once the model is trained.

Learning = turning the knobs until the machine's outputs match the data. That's it. Everything else is about *how* to turn them and *how to know* the result is any good.

Composition: feeding one machine into another. In $\ell(f(x), y)$, first f turns x into a prediction $\hat{y} = f(x)$, then ℓ compares $\hat{y}$ with the truth y. Read inside-out.

Check: If $f_\theta(x) = 2x + 1$ and $\ell(\hat{y}, y) = (\hat{y} - y)^2$, what is $\ell(f_\theta(3), 5)$? → $f(3) = 7$, $(7-5)^2 = 4$.

4 Summation Σ and averages

Intuition. Σ (capital Greek sigma, "S" for Sum) is a compact for-loop.

$$\sum_{i=1}^{n} a_i = a_1 + a_2 + \cdots + a_n$$

- Below: where the counter starts ($i = 1$).
- Above: where it stops (n).
- Right: what to add each time (a_i).

Examples.

$$\sum_{i=1}^{4} i = 1 + 2 + 3 + 4 = 10, \qquad \sum_{i=1}^{3} i^2 = 1 + 4 + 9 = 14, \qquad \sum_{i=1}^{5} 7 = 35.$$

Three rules you'll use constantly (they're just the distributive law):

1. Constants come out: $\sum c\, a_i = c \sum a_i$.
2. Sums split: $\sum (a_i + b_i) = \sum a_i + \sum b_i$.
3. Summing a constant n times: $\sum_{i=1}^{n} c = nc$.

The mean (average) is a sum divided by the count:

$$\bar{a} = \frac{1}{n} \sum_{i=1}^{n} a_i.$$

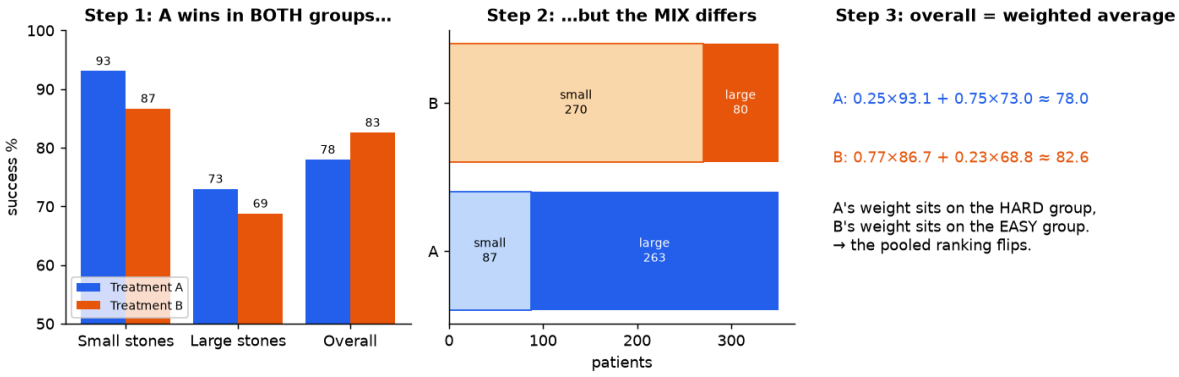
The bar $\bar{a}$ is another decoration meaning "average of".

Why ML cares. The training error of a model is literally "the average of the per-example errors":

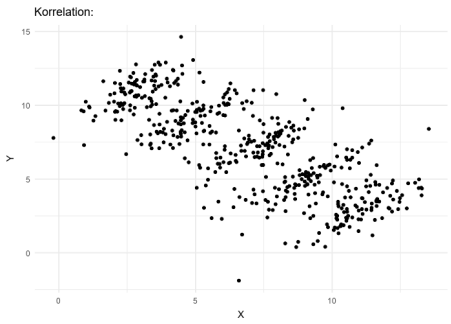
$$\hat{R} = \frac{1}{n} \sum_{i=1}^{n} \underbrace{\ell(f(x_i), y_i)}_{\text{error on example } i} .$$

Decompress: "go through every example, compute how wrong the model was, add those up, divide by how many." You can now read the most important formula of Lecture 1.

5 Weighted averages — the secret behind Simpson's paradox



Step by step: A wins inside each group, but A's patients are mostly the hard cases, so its weighted average falls below B's.



Simpson's paradox: each group trends up, the pooled data trends down. (Wikimedia Commons: Simpsons_paradox_-_animation.gif, CC BY-SA 4.0)

Intuition. An ordinary average gives every item equal say. A weighted average lets some items count more.

$$\text{weighted average} = \sum_k w_k\, a_k, \qquad \text{with } w_k \geq 0,\ \sum_k w_k = 1.$$

Example. Your marks: 90 in a test worth 20%, 60 in an exam worth 80%. Overall $= 0.2 \times 90 + 0.8 \times 60 = 66$, not the plain average 75. The exam dominates because it has more weight.

Key fact: a weighted average always lies between the smallest and largest a_k, and it is pulled toward whichever a_k has the biggest weight.

Simpson's paradox, explained by weighted averages. Kidney-stone data (Part B, Slide 14):

	Small stones	Large stones	Overall
Treatment A	93.1% (87 patients)	73.0% (263 patients)	78.0%
Treatment B	86.7% (270 patients)	68.8% (80 patients)	82.6%

Each overall rate is a weighted average of the group rates, **weighted by how many patients were in each group**:

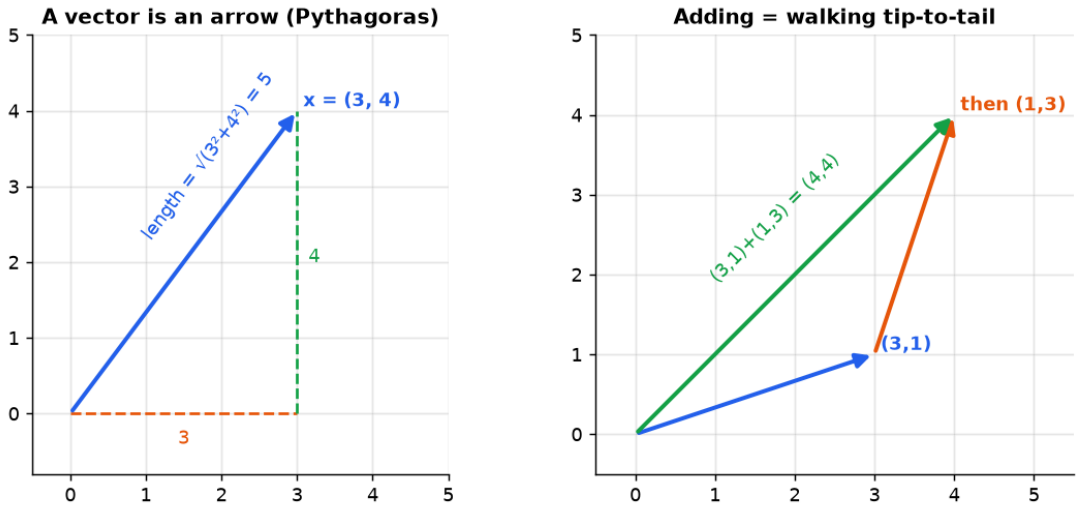
$$\text{A: } \tfrac{87}{350}(93.1\%) + \tfrac{263}{350}(73.0\%) = 0.25(93.1) + 0.75(73.0) \approx 78.0\%$$

$$\text{B: } \tfrac{270}{350}(86.7\%) + \tfrac{80}{350}(68.8\%) = 0.77(86.7) + 0.23(68.8) \approx 82.6\%$$

A beats B **inside both groups**. But A's average is dragged down because 75% of its weight sits on the hard group, while B's average is lifted because 77% of its weight sits on the easy group. **The reversal comes from the weights, not the treatments.**

This is a general law, not a trick: whenever the *mix* of groups differs between the things you compare, the pooled numbers can say the opposite of the grouped numbers.

6 Vectors — a list that is also an arrow



Left: a vector's length is Pythagoras. Right: adding vectors = walking one arrow, then the other.

Intuition 1: a list. A vector is an ordered list of numbers describing one thing:

$$x = \begin{pmatrix} 120 \\ 45 \\ 0.3 \end{pmatrix} \quad \leftarrow \text{rainfall (mm)}, \text{catchment area (km}^2\text{)}, \text{slope}$$

$x \in \mathbb{R}^3$ because it has 3 real entries. Entry j is written x_j.

Intuition 2: an arrow. In $\mathbb{R}^2$, the vector $(3, 4)$ is an arrow from the origin to the point $(3, 4)$. Its length (Pythagoras, class 10!) is $\sqrt{3^2 + 4^2} = 5$. Each example in a dataset is a **point in feature space**; similar examples are nearby points.

Operations (entry by entry):

- Add: $(1, 2) + (3, 5) = (4, 7)$.
- Scale: $3 \cdot (1, 2) = (3, 6)$.

The dot product — the most important operation in ML. Multiply matching entries and add:

$$w^\top x = w_1 x_1 + w_2 x_2 + \cdots + w_d x_d = \sum_{j=1}^{d} w_j x_j.$$

Example: $w = (0.02,\ 0.01,\ 2)$, $x = (120,\ 45,\ 0.3)$:

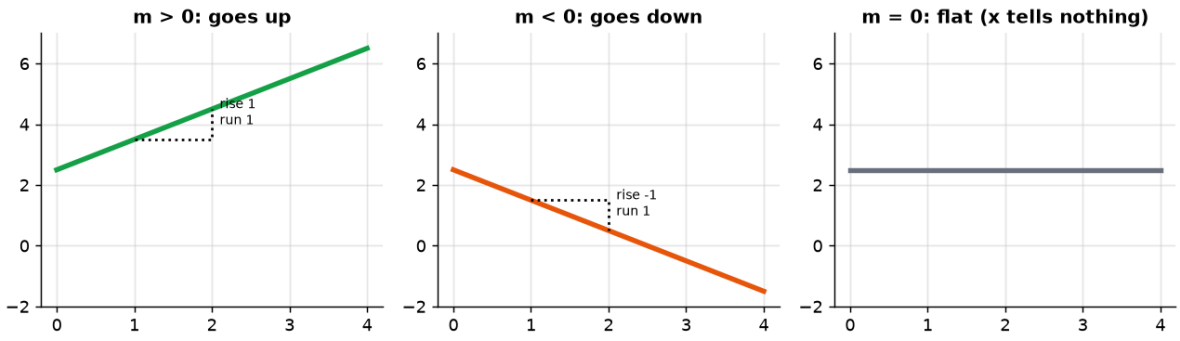
$$w^\top x = 0.02(120) + 0.01(45) + 2(0.3) = 2.4 + 0.45 + 0.6 = 3.45.$$

Interpretation: a dot product is a **weighted vote**. Each feature votes, and w_j says how much that feature's vote counts (and in which direction, if negative). A **linear model** is just

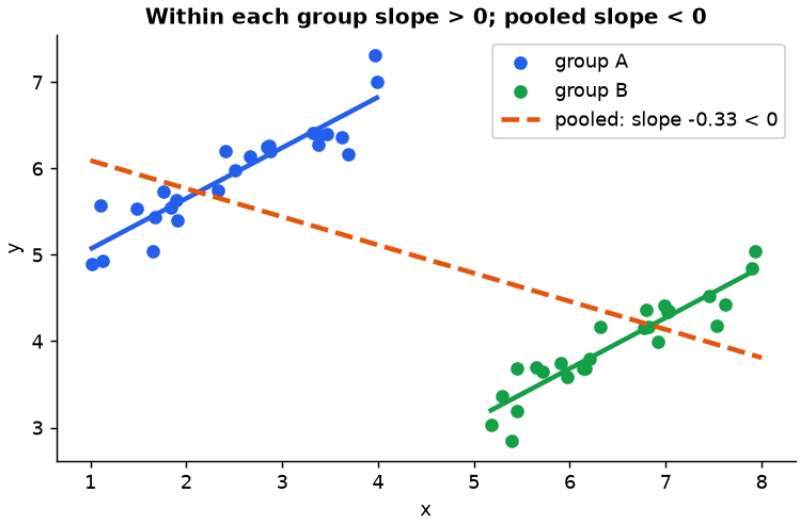
$$\hat{y} = w^\top x + b,$$

i.e. $y = mx + c$ upgraded to many inputs. (The $\top$, "transpose", is explained properly in the Lecture 2 file; for now read $w^\top x$ as "dot product of w and x".)

7 Lines, slopes, and trends



The sign of m is all that 'positive trend' or 'negative trend' means.



The §7 picture: both groups slope up, the single line through all points slopes down.

From coordinate geometry: the line $y = mx + c$ has **slope** $m = \frac{\Delta y}{\Delta x}$ ("rise over run").

- $m > 0$: going right, the line goes **up** (positive trend).
- $m < 0$: going right, it goes **down** (negative trend).
- $m = 0$: flat; x tells you nothing about y.

Trend line through data. Real data doesn't sit on a line, but we can ask for the line that passes "closest" to all the points. Its slope summarises the relationship. (How to find that line is Lecture 2–4: least squares.)

Simpson's paradox, in pictures (Lecture 1, slide 14). Imagine:

- Group A: points at $x \in [1, 4]$, y rising from 5 to 7.
- Group B: points at $x \in [5, 8]$, y rising from 3 to 5.

Inside each group, slope > 0. But B sits to the right and lower than A. A single line through *all* points must go from A's high cloud down to B's low cloud: slope < 0. Same phenomenon as §5: the pooled trend is dominated by the **difference between groups**, not the trend within them.

8 Probability, rebuilt properly

Class 10 gave you: $P(E) = \dfrac{\text{favourable outcomes}}{\text{total outcomes}}$ for equally likely outcomes. Let's extend that into the toolkit ML actually uses.

8.1 The basic objects

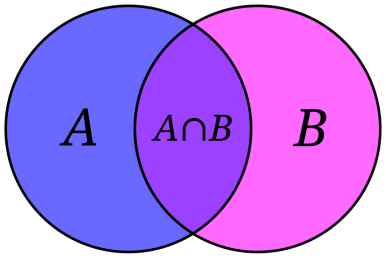
- **Experiment**: something with an uncertain result (roll a die, observe tomorrow's rainfall).
- **Sample space** Ω: set of all possible results. Die: $\Omega = \{1, \ldots, 6\}$.
- **Event**: a subset of Ω. "Even" $= \{2, 4, 6\}$.
- **Probability** $P(E)$: a number in $[0, 1]$ measuring how likely E is.

The three rules everything follows from:

1. $0 \le P(E) \le 1$.
2. $P(\Omega) = 1$ (something happens).
3. If E and F can't both happen, $P(E \text{ or } F) = P(E) + P(F)$.

Consequence: $P(\text{not } E) = 1 - P(E)$.

8.2 Joint probability — "and"



A ∩ B: outcomes in both A and B. Joint probability is the size of this overlap. (Wikimedia Commons: Venn_A_intersect_B.svg, Public domain)

$P(A, B)$ (also written $P(A \cap B)$) is the probability that **both** happen.

Use a two-way table. 100 days of records:

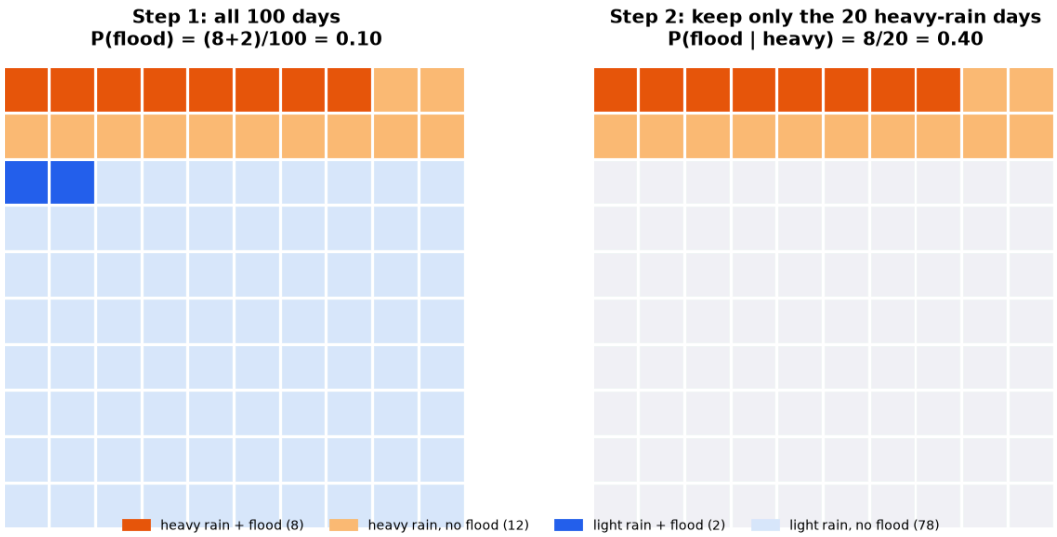
	Flood	No flood	Total
Heavy rain	8	12	20
Light rain	2	78	80
Total	10	90	100

$P(\text{heavy}, \text{flood}) = 8/100 = 0.08.$

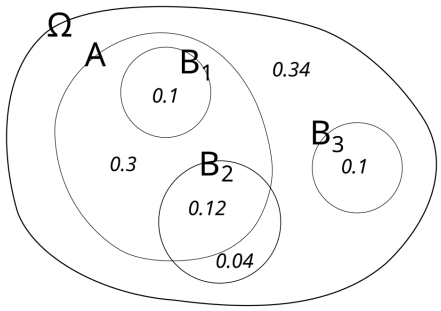
8.3 Marginal probability — "ignore the other variable"

Add across a row or column: $P(\text{flood}) = (8 + 2)/100 = 0.10$. It's called *marginal* because you write it in the table's margin.

8.4 Conditional probability — "shrink the universe"



Conditioning = throwing away the squares that don't match what you know, then counting again.



Shrink the universe to B, then ask what share of B is also A. $P(A|B_2)$ = 0.12 / (0.12 + 0.04) = 0.75; $P(A|B_1)$ = 1 (B_1 sits inside A); $P(A|B_3)$ = 0 (no overlap). (Wikimedia Commons: Conditional_probability.svg, CC0)

Intuition. $P(A \mid B)$ = "probability of A **given** that we already know B happened". Knowing B throws away every outcome where B didn't happen. Your new universe is just the B row/column.

$P(\text{flood} \mid \text{heavy})$: look only at the heavy-rain row (20 days). Floods: 8. So $8/20 = 0.4$.

Formula (same thing, written with probabilities):

$$P(A \mid B) = \frac{P(A, B)}{P(B)} = \frac{0.08}{0.20} = 0.4.$$

Compare: $P(\text{flood}) = 0.1$ but $P(\text{flood} \mid \text{heavy}) = 0.4$. **Information changes probability.** That is what a predictive model exploits: it estimates $P(y \mid x)$, "how likely is each answer *given* what I see?"

Direction matters. $P(\text{heavy} \mid \text{flood}) = 8/10 = 0.8 \neq 0.4$. "Most floods had heavy rain" is not "most heavy rain causes floods".

Multiplication rule (rearrange the formula):

$$P(A, B) = P(A \mid B)\, P(B).$$

8.5 Conditioning on more than one thing

$P(Y \mid X, G)$: condition on both X and a third variable G — shrink the universe to rows where *both* hold. Simpson's paradox is exactly the statement:

$$P(Y \mid X) \text{ and } P(Y \mid X, G) \text{ can point in opposite directions.}$$

In §5: $P(\text{success} \mid \text{A}) < P(\text{success} \mid \text{B})$, yet $P(\text{success} \mid \text{A}, \text{small}) > P(\text{success} \mid \text{B}, \text{small})$ and the same for large.

8.6 Independence

A and B are **independent** if knowing one tells you nothing about the other:

$$P(A \mid B) = P(A) \quad \Longleftrightarrow \quad P(A, B) = P(A)\, P(B).$$

Two coin flips are independent. Rain and flood are not ($0.4 \neq 0.1$).

8.7 Total probability — "split into cases, then recombine"

If $B_1, B_2, \ldots$ cover every possibility without overlapping:

$$P(A) = \sum_k P(A \mid B_k)\, P(B_k).$$

This is a weighted average (§5)! The overall chance of A is the average of the in-case chances, weighted by how common each case is.

Check with the table: $P(\text{flood}) = P(\text{flood} \mid \text{heavy})P(\text{heavy}) + P(\text{flood} \mid \text{light})P(\text{light}) = 0.4(0.2) + 0.025(0.8) = 0.08 + 0.02 = 0.1$. ✓

8.8 Bayes' rule — "flip the condition"

Combine the multiplication rule both ways: $P(A \mid B)P(B) = P(A, B) = P(B \mid A)P(A)$. Divide:

$$\boxed{P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}}$$

Why it matters for ML — the rare-event trap. A disease affects 1% of people. A test catches 99% of sick people and wrongly flags 5% of healthy people. You test positive. Chance you're sick?

$$P(\text{sick} \mid +) = \frac{P(+ \mid \text{sick})P(\text{sick})}{P(+)} = \frac{0.99 \times 0.01}{0.99 \times 0.01 + 0.05 \times 0.99} = \frac{0.0099}{0.0594} \approx 0.17.$$

Only **17%**! Because healthy people are 99× more common, their 5% false alarms outnumber the real cases. The **base rate** $P(\text{sick})$ dominates. Same root cause as Lecture 1's accuracy paradox: when one class is rare, intuition built on balanced cases breaks.

9 Random variables, distributions, expectation

9.1 Random variable

A **random variable** is a number whose value is decided by chance. Convention:

- Capital X, Y: the random variable itself (tomorrow's rainfall, not yet known).
- Lowercase x, y: a particular value it took (it rained 90 mm).

9.2 Distribution

The **distribution** of X lists every possible value and how likely it is. For a fair die, $p(x) = 1/6$ for $x = 1, \ldots, 6$. Probabilities over all values sum to 1.

Lecture 1 writes $\mathcal{D}$ for **the unknown distribution that generates the data** — think of it as "the world's dice". Every real flood event, every patient, every email is one roll of these dice. We never see the dice, only the rolls.

$(X, Y) \sim \mathcal{D}$ reads "the pair (X, Y) **is drawn from** $\mathcal{D}$".

9.3 iid

A dataset is often assumed **iid** — *independent and identically distributed*:

- **Identically distributed**: every example comes from the *same* $\mathcal{D}$ (same dice).
- **Independent**: one example doesn't influence another (rolls don't affect each other).

Where it breaks: consecutive days of weather (today affects tomorrow), neighbouring towns (similar weather), many rows from the same user, data from before and after a policy change.

9.4 Expectation — the long-run average

> **Intuition.** Roll a die a million times and average the results. You'll get very close to 3.5. That limit is the **expected value**:

$$\mathbb{E}[X] = \sum_x x\, p(x).$$

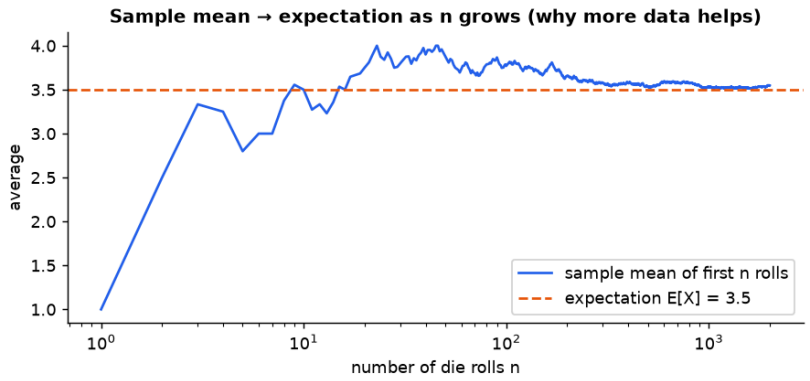
Die: $\mathbb{E}[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \cdots + 6 \cdot \frac{1}{6} = \frac{21}{6} = 3.5.$

Look closely: **expectation is a weighted average of the possible values, weighted by their probabilities.** (§5 again! Weighted averages are everywhere.)

Expectation of a *function* of X works the same way: $\mathbb{E}[g(X)] = \sum_x g(x)\, p(x)$. E.g. $\mathbb{E}[X^2]$ for a die $= \frac{1+4+9+16+25+36}{6} \approx 15.17.$

Subscripts say *what is random*: $\mathbb{E}_{(X,Y)\sim\mathcal{D}}[\cdot]$ = "average over pairs (X, Y) drawn from $\mathcal{D}$".

9.5 Expectation vs sample mean — the heart of ML



Roll a die again and again: the running average wanders, then settles at E[X] = 3.5.

	True expectation	Sample mean
Formula	$\mathbb{E}[g(X)] = \sum_x g(x)p(x)$	$\frac{1}{n}\sum_{i=1}^{n} g(x_i)$
Needs	the full distribution	only n observed values
Computable?	No (we don't know $\mathcal{D}$)	Yes

Law of large numbers (intuition): as n grows, the sample mean gets closer to the expectation. With 6 die rolls your average might be 4.2; with 6,000 it'll be about 3.5.

Now apply it to errors:

$$\underbrace{R(f) = \mathbb{E}_{(X,Y)\sim\mathcal{D}}\big[\ell(f(X), Y)\big]}_{\text{true risk: what we want}} \qquad \underbrace{\hat{R}(f) = \frac{1}{n}\sum_{i=1}^{n} \ell(f(x_i), y_i)}_{\text{empirical risk: what we can compute}}$$

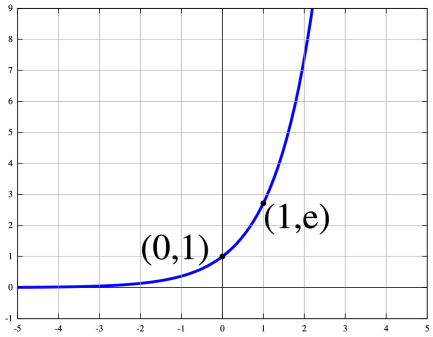
$\hat{R}$ is the sample-mean estimate of R. **But there's a catch** the law of large numbers doesn't cover: we *choose* f to make $\hat{R}$ small on these very examples. A model that memorised the training set has $\hat{R} = 0$ but may have large R. That gap is **overfitting**, and fighting it is half of this course.

10 Exponents → e → logarithms

10.1 Exponents (class 9–10 recap)

a^n = a multiplied n times. Rules: $a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, $a^0 = 1$, $a^{-n} = 1/a^n$.

10.2 The number e



y = eˣ: always positive, slope at each point equals its height. (Wikimedia Commons: Exp.svg, CC BY-SA 3.0)

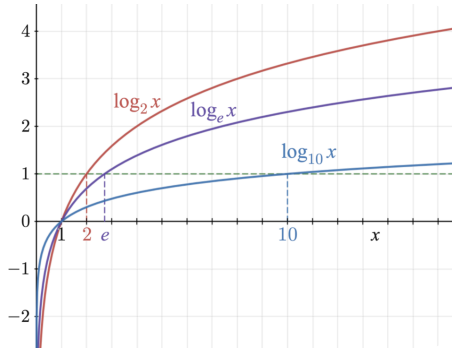
Put ₹1 in a bank at 100% interest per year.

- Paid once a year: $(1+1)^1 = 2$.
- Twice: $(1+\frac{1}{2})^2 = 2.25$.
- Monthly: $(1+\frac{1}{12})^{12} \approx 2.613$.
- Every instant: $\lim_{n\to\infty}(1+\frac{1}{n})^n = e \approx 2.71828$.

e is the natural base of continuous growth. e^z (also written $\exp(z)$):

- is always positive,
- $e^0 = 1$,
- grows explosively for $z > 0$, shrinks toward 0 for $z < 0$.

10.3 Logarithms — the inverse of exponents



log curves for several bases: all pass through (1, 0) and grow slowly. (Wikimedia Commons: Logarithm_plots.png, CC BY-SA 3.0)

Intuition. $\log_b(y)$ answers: "**what power** of b gives y?"

$$\log_{10}(1000) = 3 \text{ because } 10^3 = 1000; \qquad \log_2(8) = 3.$$

In ML, $\log$ means the **natural log** $\ln = \log_e$, so $\log(e^z) = z$ and $e^{\log y} = y$.

Rules (each is an exponent rule read backwards):

Log rule	Comes from
$\log(ab) = \log a + \log b$	$e^m e^n = e^{m+n}$
$\log(a/b) = \log a - \log b$	
$\log(a^k) = k \log a$	
$\log 1 = 0$	$e^0 = 1$

Shape. $\log y$ is only defined for $y > 0$. It's increasing (bigger input → bigger output). It's negative for $0 < y < 1$, zero at $y = 1$, and plunges to $-\infty$ as $y \to 0$.

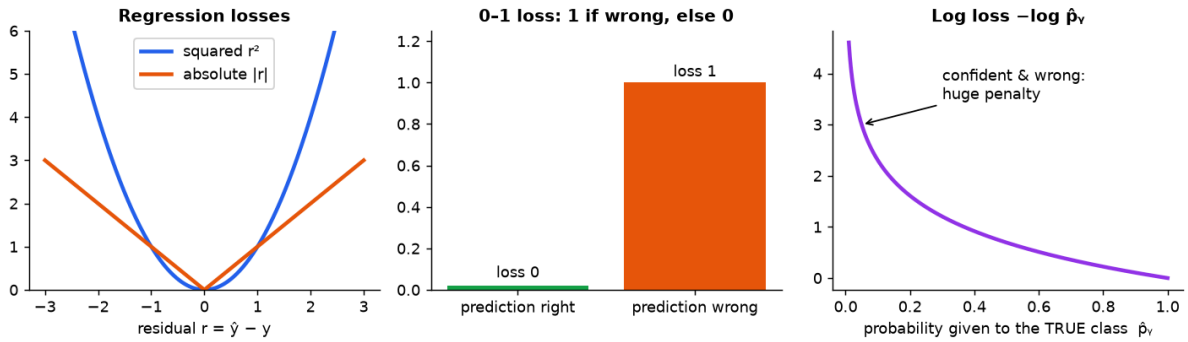
10.4 Why ML loves logs

1. **Products → sums.** Probabilities of independent events multiply. Multiplying 1000 numbers like 0.01 gives 10^{-2000} — a computer rounds that to 0. Taking logs turns it into a sum of 1000 numbers like -4.6: perfectly manageable. (Central in Lecture 2.)
2. **Log preserves order.** Because $\log$ is increasing, whatever maximises P also maximises $\log P$. So we can switch to logs freely when searching for the best model.
3. **$-\log p$ is a natural "surprise" score.**

Model's probability for the true answer, p	$-\log p$	Feeling
0.99	0.01	"I knew it"
0.9	0.11	
0.5	0.69	coin flip
0.1	2.30	surprised
0.01	4.61	shocked
→ 0	→ ∞	"impossible!"

That's the **log loss** from Lecture 1: penalty $= -\log \hat{p}_y$.

11 Squares, absolute values, indicators — ways to measure "wrong"



Three ways to measure 'wrong': squared grows fast, absolute grows steadily, log loss explodes when you're confident and wrong.

Let the **residual** be $r = y - \hat{y}$ (truth minus prediction). It can be positive or negative, so we can't just average residuals — $+3$ and -3 would cancel to "zero error". Three fixes:

Absolute value $|r|$: drop the sign. Errors of 1 and 3 cost 1 and 3. Penalty grows *linearly*.

Square r^2: also drops the sign, but grows *quadratically*. Errors of 1 and 3 cost 1 and **9**. Big mistakes are punished much more. Also smooth (no sharp corner at 0), which makes calculus easy — that's why it's the default for regression. (Lecture 2 shows a deeper reason: it comes from assuming bell-curve noise.)

Indicator $1\{\text{condition}\}$: 1 if true, 0 if false.

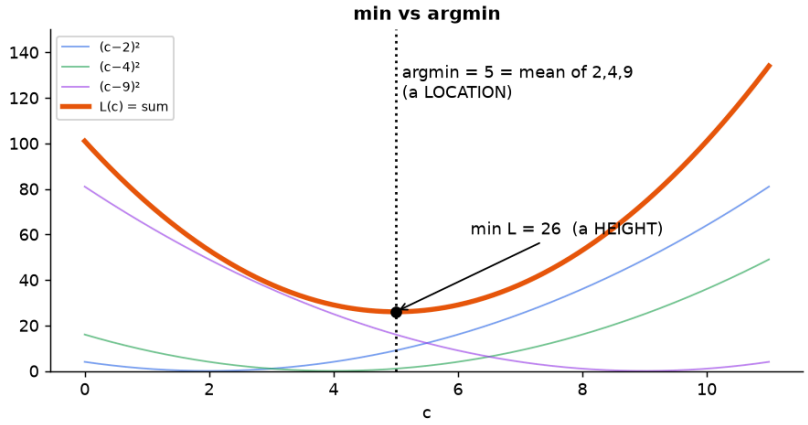
$$1\{\hat{y} \neq y\} = \begin{cases} 1 & \text{wrong} \\ 0 & \text{right} \end{cases}$$

That's the **0-1 loss**. It only asks "right or wrong?", not "how wrong?". Averaging it gives the error rate:

$$\frac{1}{n}\sum_{i=1}^{n} 1\{\hat{y}_i \neq y_i\} = \frac{\#\text{mistakes}}{n} = 1 - \text{accuracy}.$$

The general idea: a **loss function** $\ell(\hat{y}, y)$ is any rule that turns (prediction, truth) into a non-negative "how bad" number. **Choosing it means choosing which mistakes matter.**

12 min, max, argmin — finding the best



The mini-problem 2, 4, 9: add three parabolas; the lowest point is at c = 5 (argmin) with height 26 (min).

For a function $L(u)$:

- $\min_u L(u)$ — the **lowest value** L reaches. (A number on the *vertical* axis.)
- $\arg\min_u L(u)$ — the **input** u where that happens. (A location on the *horizontal* axis.)
- $\max$, $\arg\max$ — same for the highest.

Class 10 tool: completing the square. Any quadratic $L(u) = a(u-h)^2 + k$ with $a > 0$ has a lowest point (vertex) at $u = h$, value k, because a square is never negative.

$$L(u) = (u-3)^2 \Rightarrow \min_u L = 0, \quad \arg\min_u L = 3.$$

$$L(u) = u^2 - 6u + 11 = (u-3)^2 + 2 \Rightarrow \min = 2, \quad \arg\min = 3.$$

General: $au^2 + bu + c$ with $a > 0$ has $\arg\min = -\dfrac{b}{2a}$.

Mini-ML problem you can already solve. You have three measurements of the same quantity: 2, 4, 9. Which single number c is "closest" to all of them, measured by total squared error?

$$L(c) = (c-2)^2 + (c-4)^2 + (c-9)^2 = 3c^2 - 30c + 101.$$

$\arg\min = -(-30)/(2 \cdot 3) = 5$ — **the mean** $(2+4+9)/3$! Fact worth remembering: *the mean is the best constant predictor under squared loss.* (Under absolute loss it's the median — 4 here.)

Why ML cares. Training = $\arg\min$ over the knobs:

$$\hat{\theta} = \arg\min_{\theta} \hat{R}(\theta).$$

We want the *knob setting* (argmin), not the error value (min). For one knob, completing the square works; for many knobs we'll need calculus (see the Lecture 2 file).

13 Rates and the confusion matrix

For yes/no prediction, every case falls in one of four boxes:

	Predicted 1	Predicted 0
Actually 1	TP (true positive)	FN (false negative — missed)
Actually 0	FP (false positive — false alarm)	TN (true negative)

Each metric is just a fraction — but **with a different denominator**, i.e. a different "universe" (§8.4: it's conditional probability!):

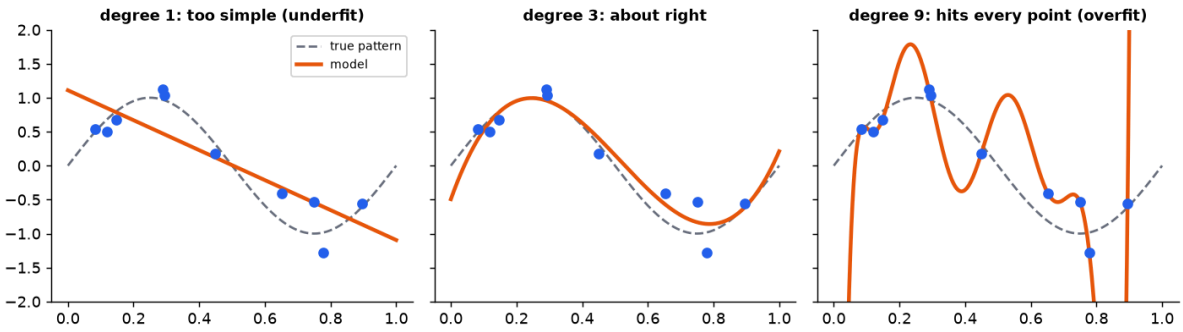
Metric	Formula	Question it answers	As a conditional probability
Accuracy	$\frac{TP+TN}{\text{all}}$	How often right overall?	$P(\hat{y} = y)$
Recall	$\frac{TP}{TP+FN}$	Of real positives, how many caught?	$P(\hat{y} = 1 \mid y = 1)$
Precision	$\frac{TP}{TP+FP}$	Of alarms raised, how many real?	$P(y = 1 \mid \hat{y} = 1)$

Lecture 1's example: 990 negatives, 10 positives, model always says 0. TP = 0, FN = 10, FP = 0, TN = 990.

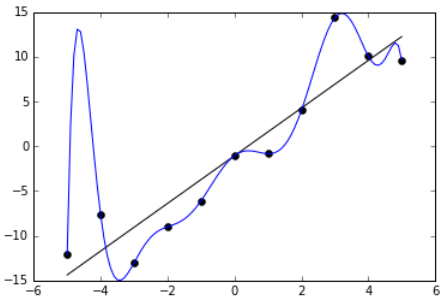
- Accuracy $= 990/1000 = 99\%$.
- Recall $= 0/10 = 0\%$.
- Precision $= 0/0$ — undefined: the model never raised an alarm.

Accuracy is a weighted average of how well you do on each class, weighted by class frequency (§5 again). If one class is 99% of the data, it gets 99% of the weight, and you can ignore the other class entirely and still score 99%.

14 Fitting curves through points — why bigger models get weird



Same 10 points, three capacities: too simple, about right, memorises the noise.



A high-degree polynomial passes through every point (overfits); the line captures the trend. (Wikimedia Commons: Overfitted_Data.png, CC BY-SA 4.0)

A fact you can verify yourself. Through 2 points you can draw exactly one line ($y = mx + c$, 2 knobs). Through 3 points (not on a line) you can draw exactly one parabola ($y = ax^2 + bx + c$, 3 knobs). In general:

> A polynomial with n knobs (degree $n - 1$) can pass **exactly** through any n points with different x values.

Capacity = how many shapes a model family can make; roughly, how many knobs.

Now take 10 noisy data points:

- **Degree 1 (a line, 2 knobs):** too stiff, misses the curve → **underfit**. Training and test errors both high.
- **Degree 3:** bends enough to follow the real shape → good test error.
- **Degree 9 (10 knobs = 10 points):** passes through *every* point exactly — training error **zero**. This is the **interpolation threshold**. But to hit every noisy point it must wiggle wildly between them → test error

explodes. **Overfit.**

That gives the classical **U-shaped** test-error curve.

Double descent (Lecture 1, slide 20). What if we use *far more* knobs than points (say 1000 knobs, 10 points)? Now infinitely many curves pass through every point. Training methods tend to pick the *smoothest* of them, and smooth curves generalise well — so test error **falls again**. The worst place is right *at* the threshold, where exactly one wiggly curve fits and the model has no freedom to be smooth.

Lesson: neither "small models are safe" nor "big models are better" is a law. Only measuring on held-out data tells you.

15 Decoding every Lecture 1 formula

Formula	Decompressed into a sentence
$S_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{n}$	"The training set is n examples, each an input paired with its correct answer."
$f : \mathcal{X} \to \mathcal{Y}$	"The model is a machine turning any possible input into an output."
f_θ	"A model whose behaviour is set by knobs θ."
$\hat{y} = f(x)$	"The prediction is what the model outputs for input x."
$p(Y \mid X)$ vs $p(Y \mid X, G)$	"The chance of outcome Y knowing only X, versus knowing X and the group G — can disagree (Simpson)."
$\ell(f(x), y)$	"How bad the model's prediction for x was, compared to truth y."
$(\hat{y} - y)^2$	"Squared miss — big misses cost a lot."
$1\{\hat{y} \neq y\}$	"1 if wrong, 0 if right."
$-\log \hat{p}_y$	"Surprise at the true answer: small if the model gave it high probability."
$\mathcal{D}$	"The unknown world process that generates examples."
$R(f) = \mathbb{E}_{(X,Y)\sim\mathcal{D}}[\ell(f(X), Y)]$	"The model's average loss over everything the world could ever produce. What we truly care about; can't compute."
$\hat{R}(f) = \frac{1}{n}\sum_i \ell(f(x_i), y_i)$	"The model's average loss on our data. Computable; an estimate of R."

If you can read this table without looking at the right column, you've got the maths of Lecture 1.

Part B Slide by Slide (all 26 pages)

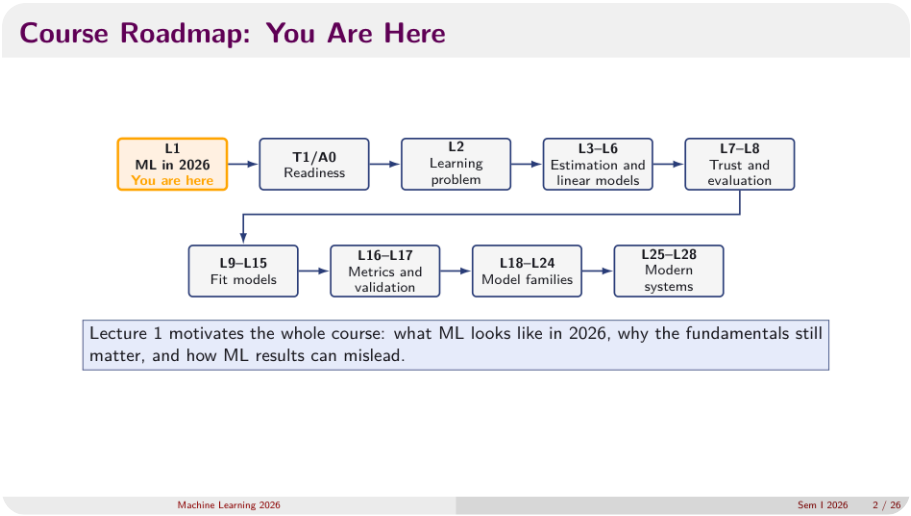
Each slide: what it says → what it means → the maths behind it (with a link back to Part A).

Slide 1 Title

Machine Learning, Lecture 1: Introduction, Past and Present. Prof. Udit Bhatia (Civil Engineering; CSE) and Prof. Krishna Prasad Miyapuram (Cognitive & Brain Sciences; AI), Semester I 2026.

Notice the instructors' backgrounds: civil engineering and brain science. That's why examples throughout (flood depth, rainfall, catchment area) come from real physical systems where wrong predictions have real costs.

Slide 2 Course Roadmap: You Are Here



Slide 2 — The orange L1 box is today; every later lecture hangs off this route.

Block	Lectures	Theme
L1	1	ML in 2026 ← **you are here**
T1/A0	tutorial	Readiness check
L2	2	The learning problem (maths language)
L3–L6	3–6	Estimation and linear models
L7–L8	7–8	Trust and evaluation
L9–L15	9–15	Fitting models
L16–L17	16–17	Metrics and validation
L18–L24	18–24	Model families
L25–L28	25–28	Modern systems

Message: Lecture 1 is motivation — *what ML looks like in 2026, why fundamentals still matter, and how ML results can mislead*. No algorithms yet; the job is to install the right suspicion.

Slide 3 Course Coordinates

- **Codes:** ES 335 (UG), ES 678 (PG).
- **Instructor:** Prof. Udit Bhatia. **Tutor:** Prof. Krishna Prasad Miyapuram.
- **Lectures:** J1 Tuesday and K2 Thursday, 14:00–15:20, room 10/103.
- **Tutorial:** B2 Friday 08:30–09:50, Jagdish Patel / Harsha Patel Learning Theatre.
- Bring the dedicated course notebook to every lecture and tutorial; changes are announced on the official channel.

Slide 4 Course Content

Four pillars, plus assessment:

1. **Foundations** — data, loss, risk; maths, probability, optimization. (Part A of these notes and the Lecture 2 file.)
2. **Supervised ML** — regression (predict a number), classification (predict a category), regularization (stop overfitting), validation (honest testing).

3. **Classical models** — decision trees, ensembles (many models voting), KNN (predict from nearest neighbours), SVMs (widest-margin boundary), PCA (compress features), clustering (group unlabeled data).
4. **Modern ML systems** — neural nets, transformers (the architecture behind LLMs), agents, accountability.

Plus assignments, quizzes, and a project.

The spine: *learn from data → evaluate honestly → reason about deployment.* Every topic hangs off this.

Slide 5 Lecture Objectives

By the end you should be able to:

1. Explain why 2026 ML still depends on classical foundations. → Slides 6–10
2. Describe the ML loop: data → representation → model → decision. → Slide 11
3. Recognise four ways results mislead: **aggregation, metrics, shortcuts, model capacity**. → Slides 13–21
4. State the notation $S_{\text{train}}, x, y, f, \ell, R, \hat{R}$. → Slides 23–26 + Part A §15
5. Understand course expectations (assignments, AI tools, quizzes, conduct). → see `course-documents/` PDFs

Guiding question: *When a model looks impressive, what would make the result untrustworthy?* Keep this question in your head for the whole course.

Slide 6 The Claim for This Course

> In 2026, ML is not just "fit a model". It is the craft of building **data-driven systems** whose predictions, explanations, and decisions remain **reliable outside the notebook**.

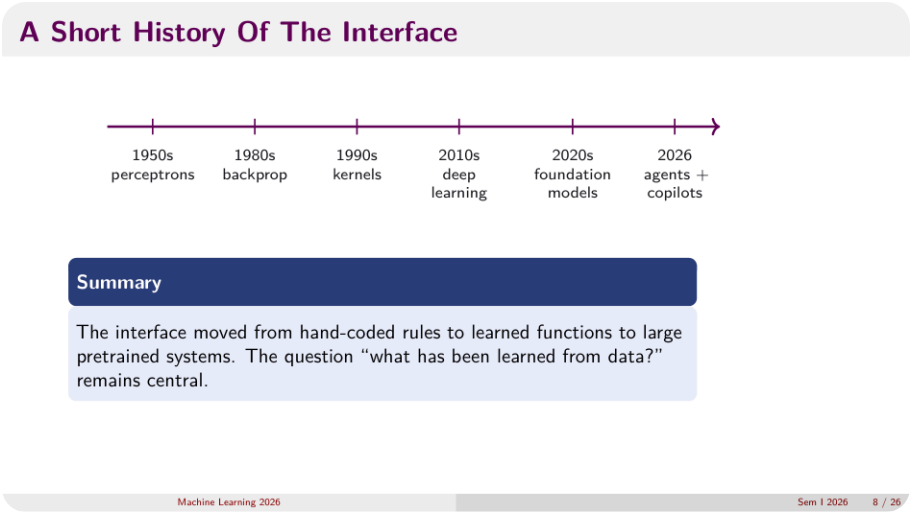
"Outside the notebook" = on new data, in the real world, after deployment. In maths language: we care about the true risk R, not the training error $\hat{R}$ (Part A §9.5).

> Insight: tools change quickly; the fundamentals of **data, loss, risk, generalization, evaluation** stay useful. A library from 2026 will be outdated by 2030; the idea "measure on held-out data" won't.

Slide 7 Section: ML in 2026

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Slide 8 A Short History of the Interface

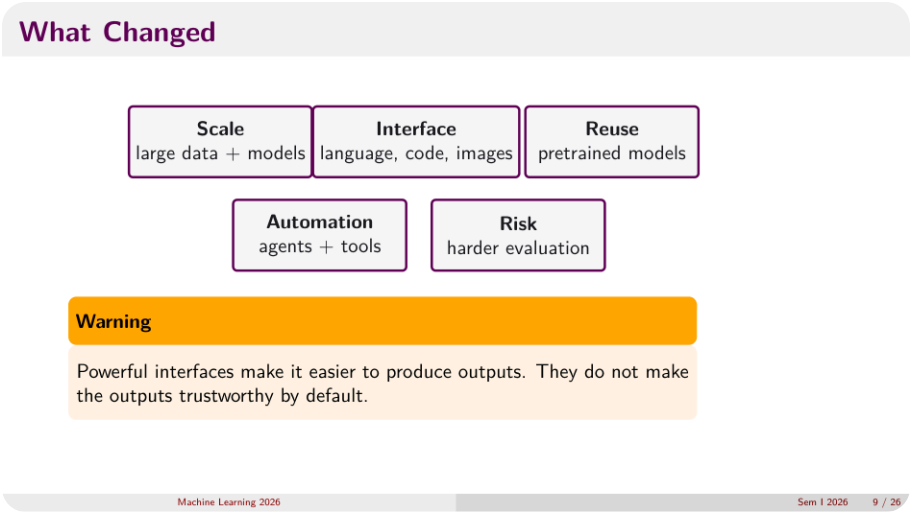


Slide 8 — Read left to right: each era hands more of the job from hand-written rules to learned functions.

Era	Milestone	Plain explanation
1950s	**Perceptron**	Compute $w^\top x + b$ (Part A §6); output 1 if positive, else 0. When it's wrong, nudge w toward the right answer. The first machine that *learned* its weights. Only able to draw straight-line boundaries.
1980s	**Backpropagation**	Stack many perceptron-like units in layers. Backprop uses the chain rule of calculus to work out how much each weight contributed to the error, so all layers can be trained together.
1990s	**Kernels** (SVMs)	Get curved boundaries from linear methods by implicitly mapping data into a richer space where a straight cut works. Strong mathematical guarantees.
2010s	**Deep learning**	Big data + GPUs make very deep networks trainable. They learn their *own* features (edges → shapes → objects) instead of humans hand-designing them.
2020s	**Foundation models**	One enormous model pretrained on huge data (most of the internet), then reused for many tasks. LLMs are the famous example.
2026	**Agents + copilots**	Models that use tools, write and run code, and take multi-step actions.

Summary: the interface moved *hand-coded rules* → *learned functions* → *large pretrained systems*. The question **"what has been learned from data?"** stays central. A 2026 agent is still a function fitted to data and inherits every problem that implies.

Slide 9 What Changed



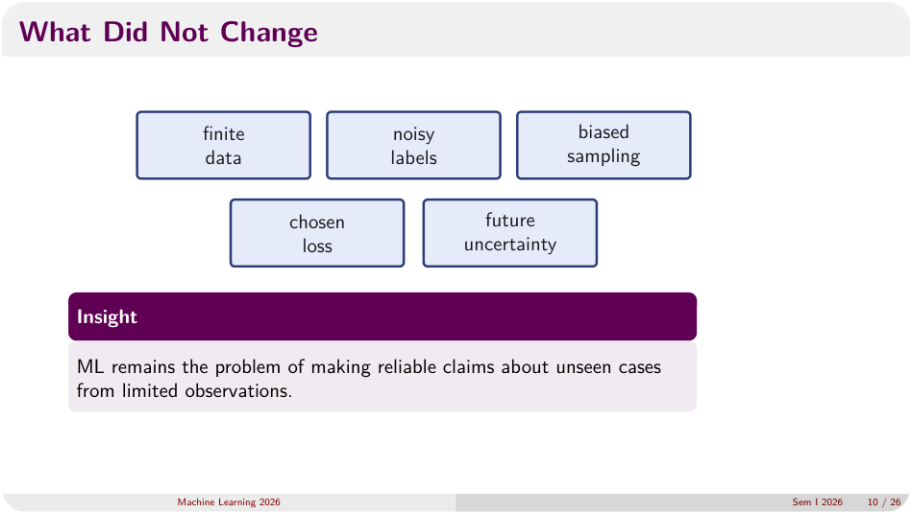
Slide 9 — Four boxes are gains (scale, interface, reuse, automation); the fifth (risk) is their price.

Change	Meaning
Scale	Datasets of trillions of words; models with billions of parameters.
Interface	You interact through language, code, images — not just tables.
Reuse	Start from a pretrained model and adapt, instead of training from scratch.
Automation	Agents chain tools: search → code → run → report.
Risk	Evaluation is **harder**: outputs are open-ended, failures are subtle, training data is unknown.

Careful. Powerful interfaces make it easier to **produce** outputs. They do not make outputs **trustworthy** by default.

A fluent, confident answer is not evidence of a correct one.

Slide 10 What Did Not Change



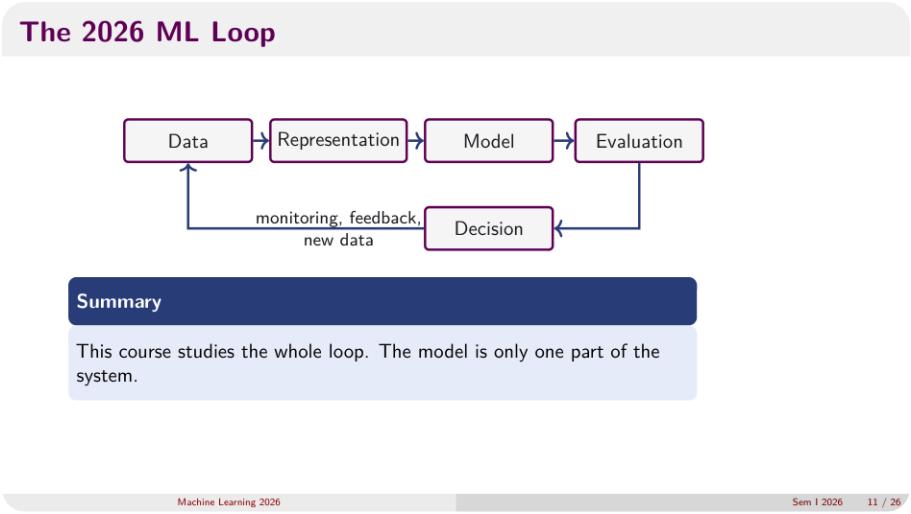
Slide 10 — Five limits that no amount of compute removes.

Constant	Meaning	Maths link
Finite data	You only ever have n examples.	$\hat{R}$ is an average over n, not the true R (§9.5)
Noisy labels	The "correct answers" contain errors.	y itself is random (§9.1)
Biased sampling	Your data isn't a fair picture of the world.	S_{train} is not a fair draw from $\mathcal{D}$ (§9.3)
Chosen loss	Someone decides what counts as "wrong".	choice of ℓ (§11)
Future uncertainty	The world shifts after you deploy.	$\mathcal{D}_{\text{past}} \neq \mathcal{D}_{\text{future}}$ — *non-stationarity*

Insight: ML remains the problem of making **reliable claims about unseen cases from limited observations.** That sentence is the definition of the field.

Slide 11

The 2026 ML Loop



Slide 11 — Follow the arrows: the decision feeds back into new data, so it's a loop, not a straight line.

```
Data ──► Representation ──► Model ──► Evaluation ──► Decision
  ▲                                                     │
  └──────────── monitoring, feedback, new data ◄────────┘
```

Stage	Question	Flood example
Data	What was collected, how, from whom?	Gauge readings — but gauges exist mostly in big cities
Representation	How do raw observations become the vector x?	rainfall, catchment area, slope → $x \in \mathbb{R}^3$
Model	Which function f, which knobs?	linear $\hat{y} = w^\top x + b$
Evaluation	How well on *unseen* cases, by which metric?	error on held-out storms
Decision	What action, at what cost?	evacuate if predicted depth > 1 m
Feedback	How do decisions change future data?	evacuated areas record fewer injuries → data looks "safer"

Summary: this course studies the *whole* loop; the model is only one part. Most real failures come from the other boxes.

Slide 12 Section: Four Paradoxes

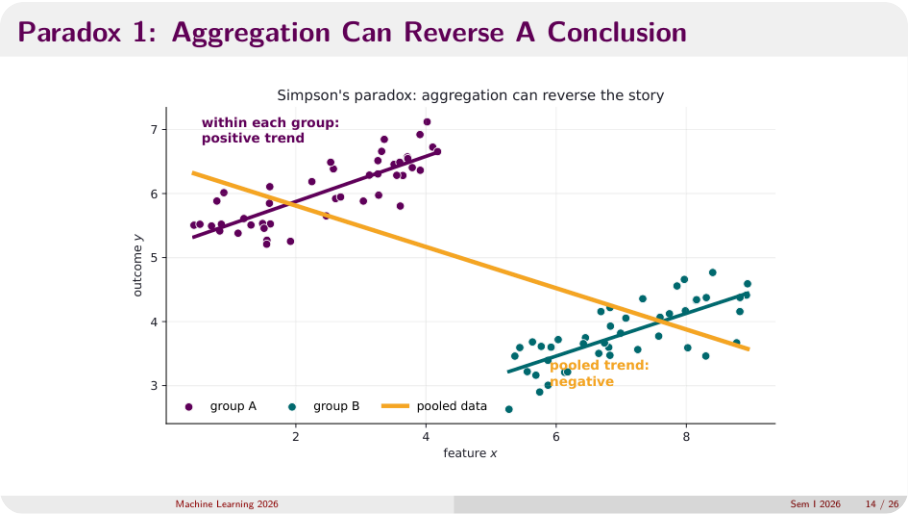
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Slide 13 Why Start With Paradoxes?

> ML is powerful partly because it finds patterns. ML is dangerous partly because it finds **the wrong patterns with great confidence**.

Question: the first skill in ML is not coding a model. It's **noticing when the apparent pattern is not the claim you think it is.** Each of the four paradoxes below is a different way a true pattern in the data supports a false claim.

Slide 14 Paradox 1: Aggregation Can Reverse a Conclusion



Slide 14 — Each coloured cloud slopes UP; the orange pooled line slopes DOWN.

The plot: horizontal axis feature x (2–8), vertical axis outcome y (3–7). Two clusters:

- **Group A** (left, higher): within it, y rises with x.
- **Group B** (right, lower): within it, y also rises with x.
- **Pooled** (both together): the best-fit line goes from A's high cloud down to B's low cloud → **negative** slope.

This is **Simpson's paradox**: aggregation reverses the story.

Why it happens (Part A §5 and §7): the pooled trend mixes two things: (1) the trend *within* each group and (2) the *gap between* groups. If the gap is large and points the other way, it wins.

Real data (kidney-stone treatments):

	Small stones	Large stones	All patients
Treatment A	81/87 = **93%**	192/263 = **73%**	273/350 = 78%
Treatment B	234/270 = 87%	55/80 = 69%	289/350 = **83%**

A wins in each group but loses overall: doctors gave A mostly to the hard (large-stone) cases. Stone size is a **confounder**: it influences both *which* treatment was given and *whether* it worked.

Slide 15 What Simpson's Paradox Teaches

Conditioning: before trusting a relationship between X and Y, ask whether a third variable G changes the story.

$$p(Y \mid X) \quad \text{may look very different from} \quad p(Y \mid X, G).$$

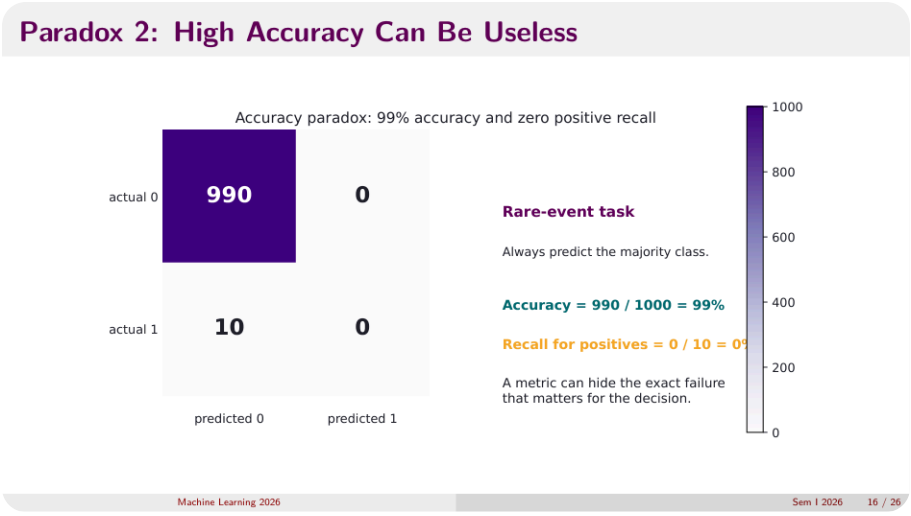
Decompress (Part A §8.4–8.5):

- $p(Y \mid X)$ — "the chance of the outcome, knowing only the treatment".
- $p(Y \mid X, G)$ — "the chance of the outcome, knowing the treatment **and** the group".

They're linked by total probability (§8.7): $p(Y \mid X) = \sum_g p(Y \mid X, g)\, p(g \mid X)$. The weights $p(g \mid X)$ — *which groups got which treatment* — differ between A and B. That's the whole paradox in one equation.

> **Insight:** subgroups, context, and sampling mechanisms are not details. They can **change the sign** of the conclusion. An ML model trained on pooled data will happily learn the reversed pattern.

Slide 16 Paradox 2: High Accuracy Can Be Useless



Slide 16 — The 990 in the top-left drowns everything; the bottom-right 0 means not one real positive was caught.

Rare-event task: 1000 cases, 990 actual 0, 10 actual 1. Strategy: *always predict the majority class (0)*.

	predicted 0	predicted 1
actual 0	990	0
actual 1	10	0

$$\text{Accuracy} = \frac{990}{1000} = 99\%, \qquad \text{Recall for positives} = \frac{0}{10} = 0\%.$$

(Bar chart on the slide: a tall bar at 990, a tiny one at 10 — the imbalance made visible.)

A metric can hide the exact failure that matters for the decision. See Part A §13: accuracy is a weighted average where the majority class gets 99% of the weight.

Slide 17 What the Accuracy Paradox Teaches

For rare-event tasks the majority class dominates accuracy. Examples where the rare class is the whole point:

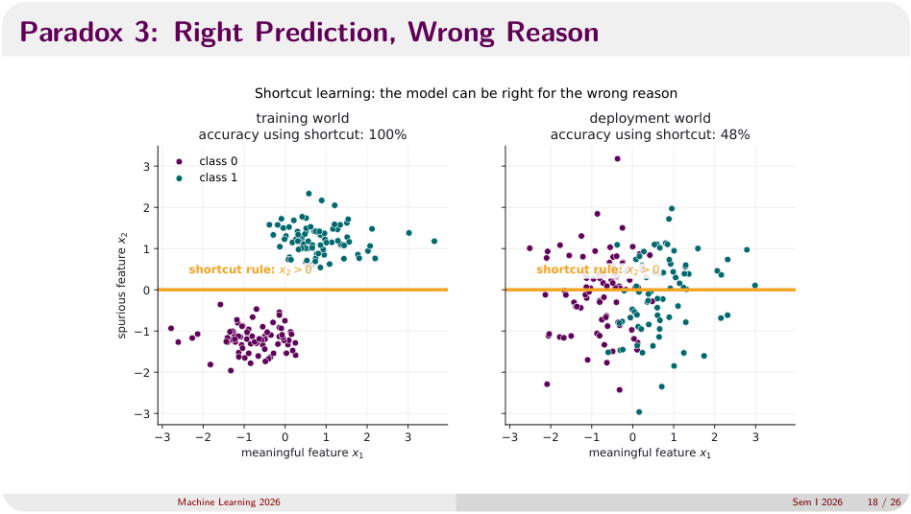
- **Flood damage detection** — most map tiles show no damage.

- **Disease screening** — most people are healthy.
- **Fraud detection** — most transactions are legitimate.
- **Equipment failure prediction** — machines usually work.

> Careful. **The metric must match the decision.** Otherwise the best score may reward the wrong behaviour.

Also notice the cost asymmetry: missing a flood (false negative) is far worse than a false alarm. Accuracy treats them as equal. Lectures 9–10 build better tools (precision, recall, ROC/AUC, cost-weighted thresholds).

Slide 18 Paradox 3: Right Prediction, Wrong Reason



Slide 18 — Left: the rule $x_2 > 0$ separates the classes perfectly. Right: the same rule at deployment is a coin flip (48%).

Two panels, each with meaningful feature x_1 (horizontal) and spurious feature x_2 (vertical), two classes.

- **Training world:** class 1 points all have $x_2 > 0$, class 0 all have $x_2 < 0$. The shortcut rule "predict 1 if $x_2 > 0$" scores **100%**.
- **Deployment world:** class depends on x_1, and x_2 is now unrelated. The same rule scores **48%** — worse than a coin flip.

Shortcut learning: the model can be right for the wrong reason. In maths terms, the model learned a pattern in $\mathcal{D}_{\text{train}}$ that does not exist in $\mathcal{D}_{\text{deploy}}$. Training-set $\hat{R}$ was perfect; deployment R is terrible.

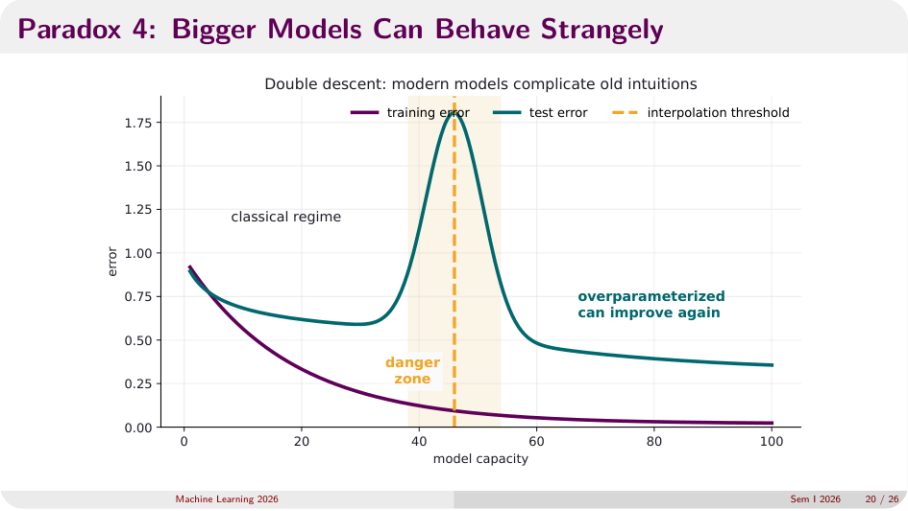
Slide 19 What Shortcut Learning Teaches

Models exploit **whatever predictive signal is easiest**. It might be the real mechanism, or a shortcut:

Shortcut	Instead of	Real example
Background	the object	"wolf" classifier actually detecting snow
Hospital	the disease	X-ray model learning which hospital's scanner (sicker patients go to certain hospitals)
Sensor artifact	the event	a gauge glitch that happens to coincide with storms
Future information	present information	a feature recorded *after* the outcome (e.g. "insurance claim filed" when predicting damage) — *data leakage*

The question to ask of every feature: *Would this feature still be available, valid, and causal enough at deployment time?*

Slide 20 Paradox 4: Bigger Models Can Behave Strangely



Slide 20 — Test error rises to a peak at the dashed interpolation line, then falls again as capacity keeps growing.

The plot: horizontal = model capacity (0–100), vertical = error.

- **Training error** (falls steadily) → reaches ~0 at the **interpolation threshold** (dashed line).

- **Test error**: falls (classical regime) → rises into a **danger zone** peaking at the threshold → then **falls again** in the **overparameterized** region.

This is **double descent**. Mechanism explained in Part A §14 (polynomials through points).

Slide 21

What Double Descent Teaches

- **Classical intuition:** more flexible models overfit → keep models small.
- **Modern twist:** after the interpolation threshold, very large models can sometimes improve again — which is partly why huge neural networks work at all.

> Insight: this does **not** remove the need for validation; it makes careful validation **more** important. When neither "small is safe" nor "big is better" holds, only measurement on held-out data can tell you where you are on the curve.

The four paradoxes together:

#	Paradox	What misleads you	Defence
1	Simpson's	Pooling hides a confounding group	Condition on relevant variables; inspect subgroups
2	Accuracy	Metric dominated by the majority class	Choose a metric matching the decision's costs
3	Shortcut	An easy spurious feature replaces the real signal	Ask whether each feature is valid at deployment
4	Double descent	Capacity-vs-error intuition breaks	Validate on held-out data

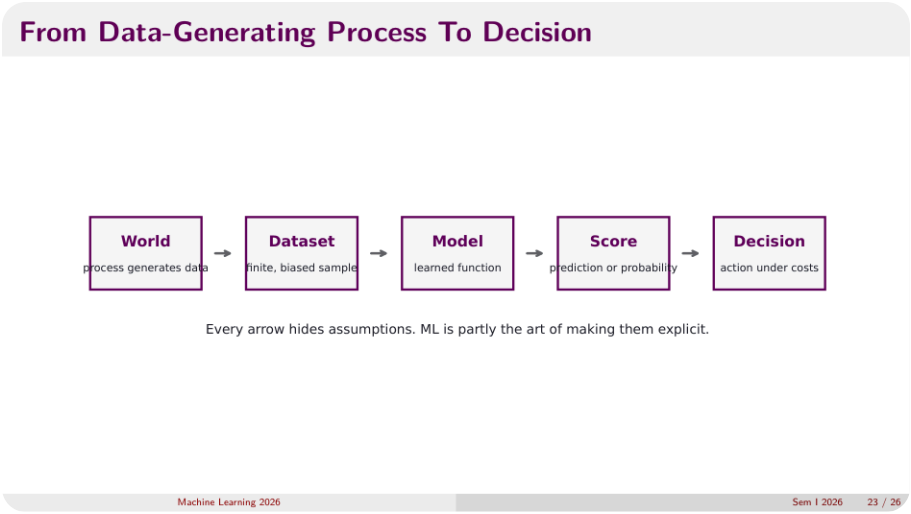
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Slide 22

Section: The Learning Language

Section divider.

Slide 23 From Data-Generating Process to Decision



Slide 23 — Every arrow hides an assumption: world → dataset → model → score → decision.

```
World ⟶ Dataset ⟶ Model ⟶ Score ⟶ Decision
```

Box	Description	Hidden assumption on the arrow *into* it
World	process generates data	—
Dataset	finite, biased sample	that sampling represents the world (iid, §9.3)
Model	learned function	that the model family can capture the pattern; that the loss is right
Score	prediction or probability	that the model works on *this* new input (no shift)
Decision	action under costs	a threshold, and the costs of each type of mistake

Every arrow hides assumptions. ML is partly the art of making them explicit.

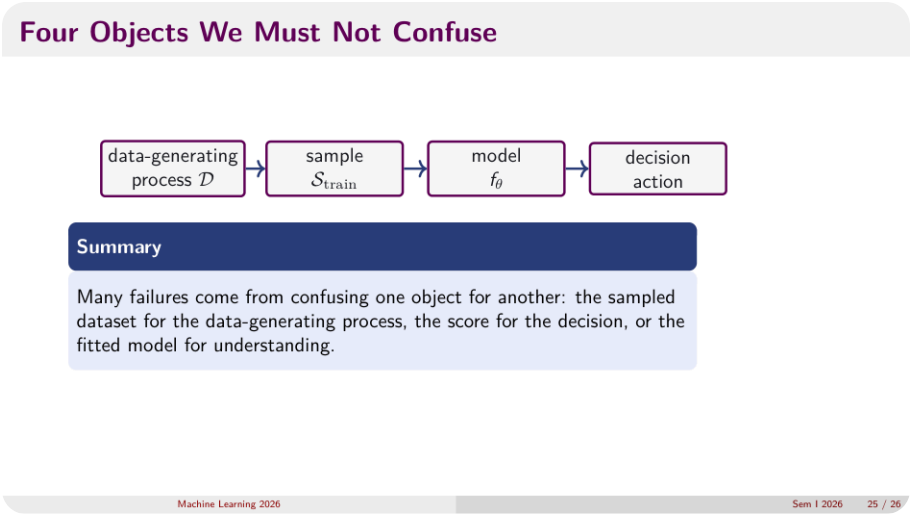
Slide 24 The Basic Learning Problem

Given examples $S_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{n}$, **learn** a function $f : \mathcal{X} \to \mathcal{Y}$ that **performs well on future examples.**

- x_i: input (features) of example i — a vector (§6).
- y_i: target (correct answer) of example i.
- f: the model (§3).

Read symbol by symbol with Part A §1–3. The crucial words are **"future examples"**: doing well on S_{train} is easy (memorise it); doing well on unseen cases — **generalization** — is the actual goal.

Slide 25 Four Objects We Must Not Confuse



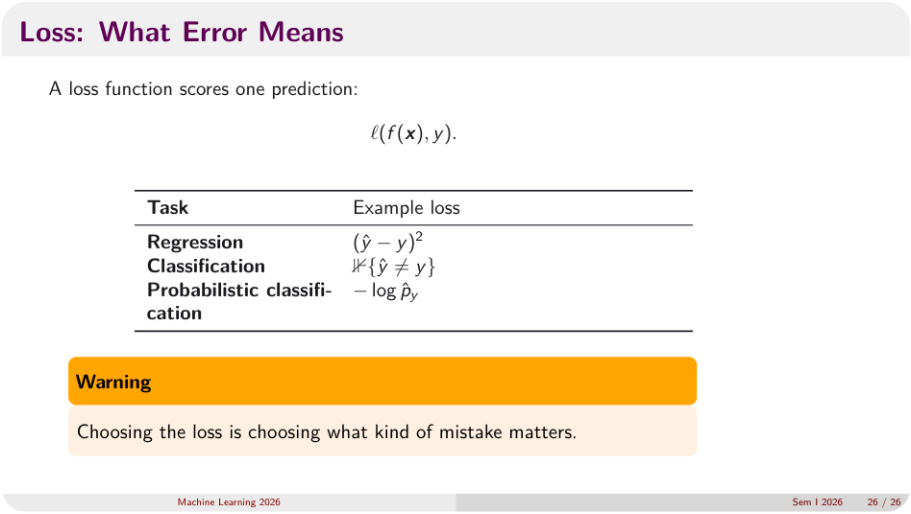
Slide 25 — Four different things: the process D, the sample, the fitted model, the action.

```
data-generating process 𝒟 ⟶ sample S_train ⟶ model f_θ ⟶ decision / action
```

Object	What it is	Typical confusion
$\mathcal{D}$	The world's unknown dice (§9.2)	—
S_{train}	n rolls of those dice	**sample ↔ world:** "my dataset says X, so the world is X"
f_θ	The fitted function with knobs θ	**fitted model ↔ understanding:** "it predicts well, so it knows *why*"
decision	The action taken	**score ↔ decision:** "probability 0.6, so act" — but acting depends on costs

Summary: many failures come from treating one of these objects as if it were another.

Slide 26 Loss: What Error Means



Slide 26 — One loss per task: squared, 0–1, and –log p.

A loss function scores **one** prediction: $\ell(f(x), y)$.

Task	Example loss	Behaviour
Regression	$(\hat{y} - y)^2$	Punishes big misses quadratically (§11)
Classification	$1\{\hat{y} \neq y\}$	Counts right/wrong only (§11)
Probabilistic classification	$-\log \hat{p}_y$	Surprise at the true class (§10.4)

Worked numbers:

- Squared: $\hat{y} = 5, y = 4 \to 1$; $\hat{y} = 7, y = 4 \to 9$.
- 0-1: wrong → 1, right → 0, no matter how close.
- Log: true class given 0.9 → 0.105; given 0.1 → 2.303.

Careful. **Choosing the loss is choosing what kind of mistake matters.**

Where the lecture ends, and what comes next. The objectives promised R and $\hat{R}$. Average the loss over the world and you get the **risk** $R(f) = \mathbb{E}_{\mathcal{D}}[\ell(f(X), Y)]$; average it over your data and you get the

empirical risk $\hat{R}(f) = \frac{1}{n}\sum_i \ell(f(x_i), y_i)$ (Part A §9.5). Training picks the f that minimises $\hat{R}$ (§12). Each paradox is a way a small $\hat{R}$ fails to mean a small R:

- Simpson / biased sampling → S_{train} doesn't represent $\mathcal{D}$.
- Accuracy paradox → the chosen ℓ doesn't match the decision.
- Shortcut → $\mathcal{D}$ at deployment differs from $\mathcal{D}$ at training.
- Double descent → the gap between $\hat{R}$ and R behaves non-intuitively with capacity.

Lecture 2 makes all of this precise.

Also worth knowing

Why error can fall twice. Past the point where a model fits every training point, many perfect fits exist. A training rule that picks the one with the smallest weights, the **minimum-norm** solution, tends to give a smooth function, so test error can fall a second time. The spike happens exactly at the threshold, where only one fit exists and it has to contort.

Part C Practice

1. Compute $\sum_{i=1}^{4}(2i-1)$. What pattern do you notice?
2. $x = (1, 0, 3)$, $w = (2, 5, -1)$, $b = 4$. Compute $\hat{y} = w^\top x + b$.
3. From the rain/flood table in §8: find $P(\text{light} \mid \text{no flood})$ and $P(\text{no flood} \mid \text{light})$.
4. Are "heavy rain" and "flood" independent? Show with numbers.
5. A fraud detector: 0.1% of transactions are fraud. It catches 95% of fraud and flags 2% of legitimate ones. Given a flag, chance of fraud?
6. Compute $\mathbb{E}[X]$ where X is 10 with probability 0.2, 0 with probability 0.5, and −4 with probability 0.3.
7. Simplify $\log(e^5 \cdot e^{-2})$ and $\log(0.2) + \log(5)$.
8. Find $\arg\min_c (c-1)^2 + (c-7)^2$. Then the min.
9. True labels $y = (1, 0, 1, 1)$, predictions $\hat{y} = (1, 1, 0, 1)$. Compute 0-1 empirical risk, accuracy, recall, precision.
0. A model gives probabilities 0.7, 0.2, 0.9 to the true classes of three examples. Compute the average log loss.
1. **Think:** You build a pass/fail predictor on students from one school and get 95% accuracy. Name two ways this could mislead you, using Lecture 1's paradoxes.

1. $1 + 3 + 5 + 7 = 16 = 4^2$. The sum of the first n odd numbers is n^2.
2. $2 + 0 - 3 + 4 = 3$.
3. $P(\text{light} \mid \text{no flood}) = 78/90 \approx 0.867$; $P(\text{no flood} \mid \text{light}) = 78/80 = 0.975$. Different denominators, different answers.
4. No: $P(\text{heavy}, \text{flood}) = 0.08$ but $P(\text{heavy})P(\text{flood}) = 0.2 \times 0.1 = 0.02$.
5. $\frac{0.95 \times 0.001}{0.95 \times 0.001 + 0.02 \times 0.999} = \frac{0.00095}{0.00095 + 0.01998} \approx 0.045$ — under 5%.
6. $10(0.2) + 0(0.5) + (-4)(0.3) = 2 - 1.2 = 0.8$.
7. $\log(e^3) = 3$; $\log(0.2 \times 5) = \log 1 = 0$.
8. Expand: $2c^2 - 16c + 50$; argmin $= 16/4 = 4$ (the mean of 1 and 7); min $= 9 + 9 = 18$.
9. Wrong on examples 2 and 3 → risk $= 2/4 = 0.5$, accuracy 0.5. TP = 2, FN = 1, FP = 1: recall $= 2/3$, precision $= 2/3$.
10. $\frac{1}{3}\left(0.357 + 1.609 + 0.105\right) \approx 0.690$.
11. Any two: (a) **shortcut** — the model may use something specific to that school (e.g. a teacher's grading quirk) that won't hold elsewhere; (b) **accuracy paradox** — if 95% of students pass,

"always predict pass" scores 95% and catches no failing students; (c) **Simpson / sampling** — one school isn't the population $\mathcal{D}$, so $\hat{R}$ there says little about R elsewhere.